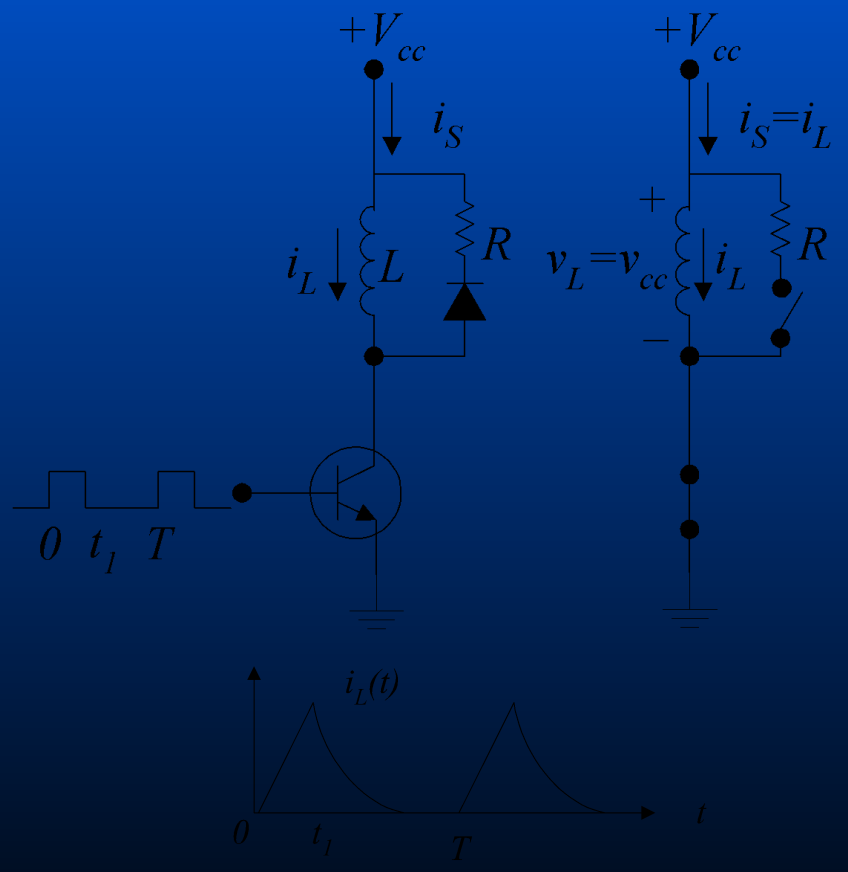
2.4 에너지 회생: Transistor On



$$0 < t < t_1$$

$$v_L = V_{CC} = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_{CC}}{L}$$

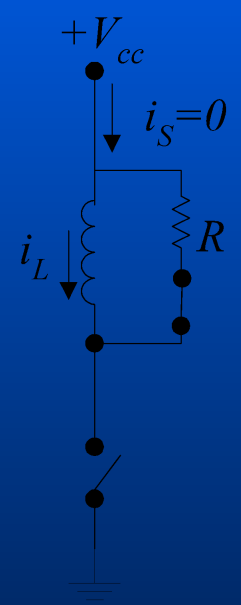
$$i_L(t) = \frac{V_{CC}}{L}\int dt + i_L(0+)$$

$$= \frac{V_{CC}}{L}\int 1dt + 0$$

$$= \frac{V_{CC}}{L}t$$

$$i_S(t) = i_L(t)$$

2.4 에너지 회생(2): Transistor Off



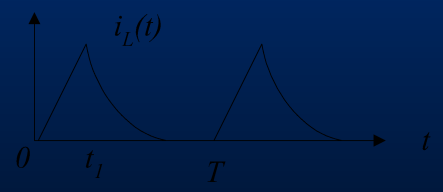
$$t_1 < t < T$$

$$L\frac{di_L}{d\lambda} + Ri_L = 0 \quad \boxed{by\ Laplace\,Transform}$$

$$L\left(SI(S) - i_0(0+)\right) + RI(S) = 0 \quad \boxed{where\ i_0(0+) = \frac{V_{CC}t_1}{L}}$$

$$SI(S) + \frac{R}{L}I(S) = \frac{V_{CC}t_1}{L},\ I(S)\left(S + \frac{R}{L}\right) = \frac{V_{CC}t_1}{L}$$

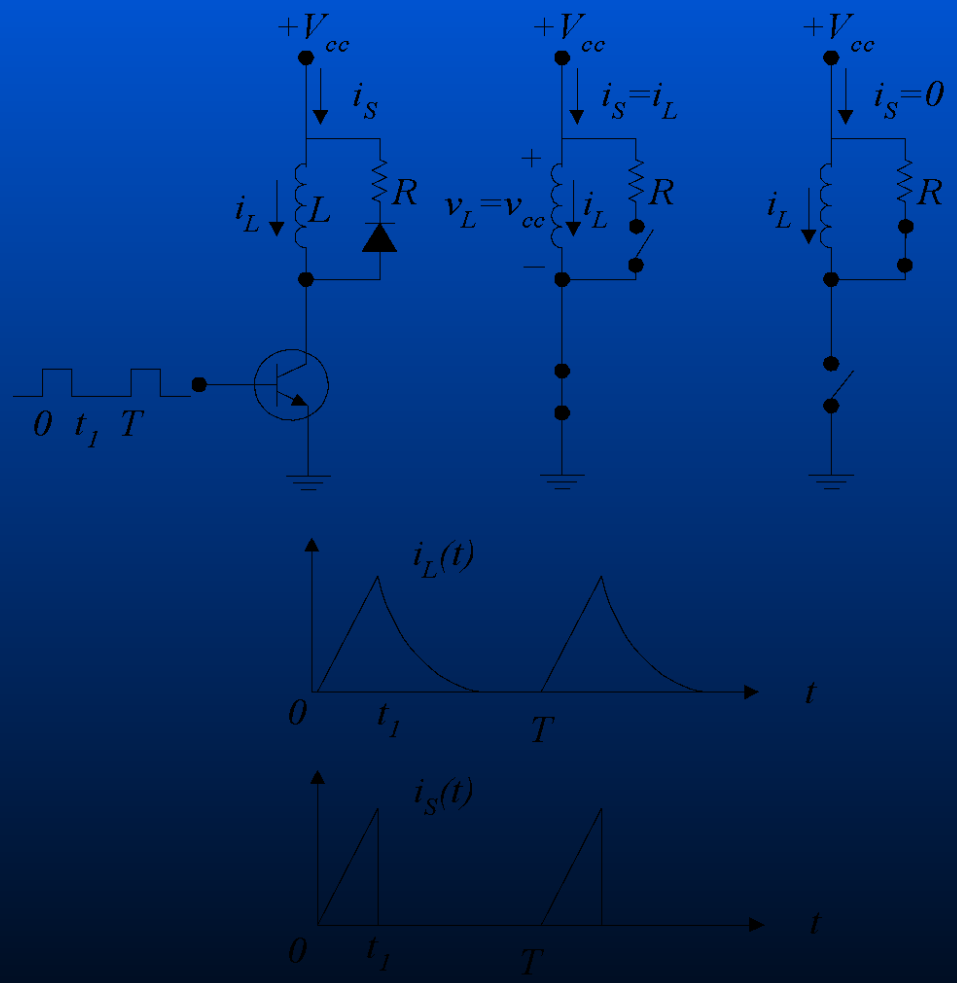
$$I(S) = \frac{V_{CC}t_1}{L}\frac{1}{\left(S + \frac{R}{L}\right)} \quad \boxed{by\ Inverse\,Laplace\,Transform}$$



$$i_L(t) = \frac{V_{CC}t_1}{L}e^{-\frac{R}{L}\lambda},\ \lambda = t - t_1, \quad i_L(t) = \frac{V_{CC}t_1}{L}e^{-\frac{R}{L}t - t_1}$$

$$i_L(t) = \frac{V_{CC}t_1}{L}e^{-\frac{(t-t_1)}{\tau}} \quad \boxed{where\ \tau = \frac{L}{R}}$$

2.4 에너지 회생(3): Summary



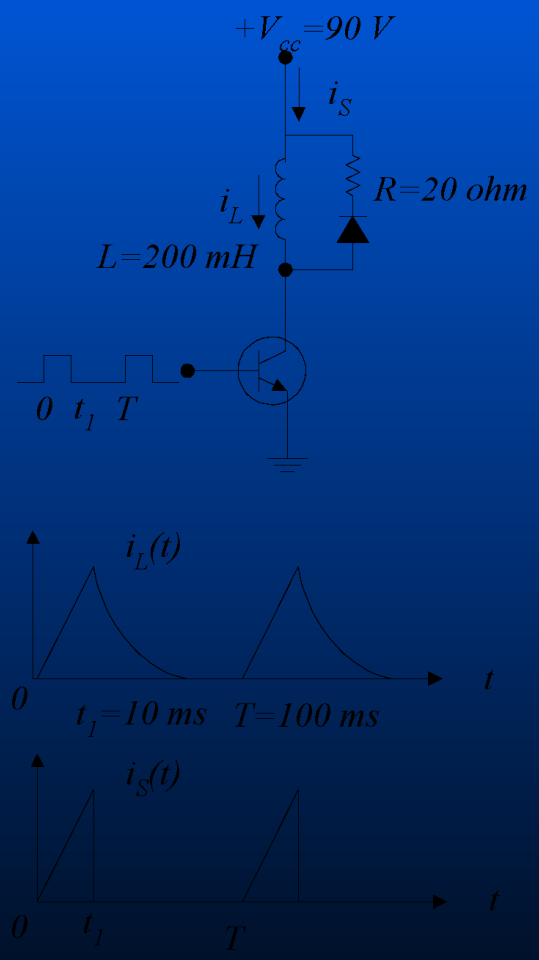
$$0 < t < t_1$$

$$i_L(t) = \frac{V_{CC} t}{L}$$

$$t_1 < t < T$$

$$i_L(t) = \frac{V_{CC} t_1}{L} e^{-\frac{(t-t_1)}{\tau}}$$

예제 2-3 회생 에너지



- 인덕터 전류

$$i_L(t) = \frac{V_{CC}t}{L} = \frac{90t}{0.2} = 450t \quad at \ 0 < t < 10mS$$

- 인덕터 첨두전류

$$i_L(t_1) = \frac{V_{CC}t_1}{L} = 450(0.01) = 4.5\,A$$

- 인덕터 에너지

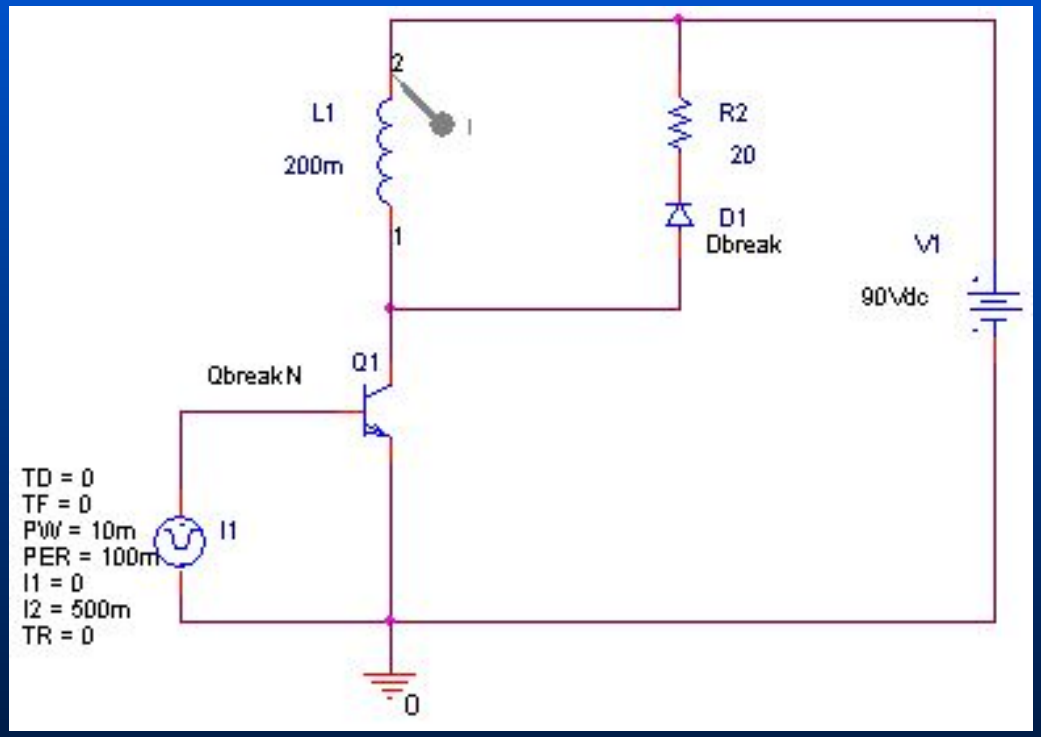
$$W_L = \frac{1}{2}Li^2(t_1) = \frac{1}{2}(0.2)(4.5)^2 = 2.025\,J$$

- 평균 소비전력

$$W_R = W_L = 2.025\,J$$

$$P_R = \frac{W_R}{T} = \frac{2.025\,J}{0.1S} = 20.25W$$

예제2-3: OrCAD Simulation

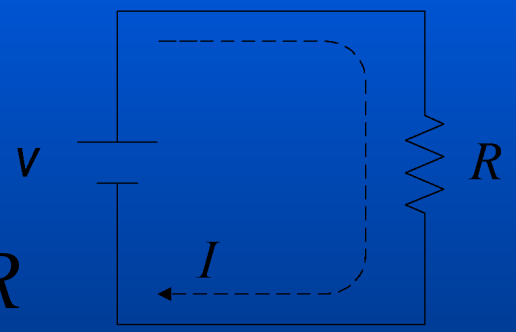


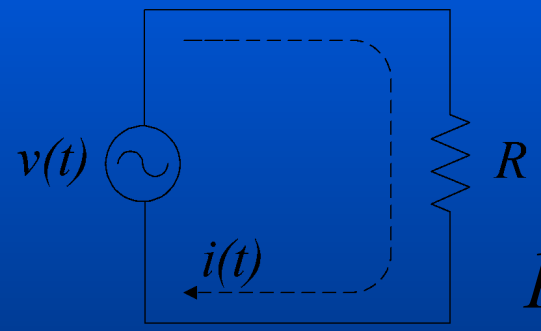
- *Parameter*
  - ✔ *R=20 Ω*
  - ✔ *L=200 mH*
  - ✔ *Ipulse*
  - ✔ *I1=1000mA*
  - ✔ *V1=90 V*
  - ✔ *Transient Step:*

    0 0.1 ms 300 ms

2.5 실효값(rms: root mean square)

$$P = I^2 R$$





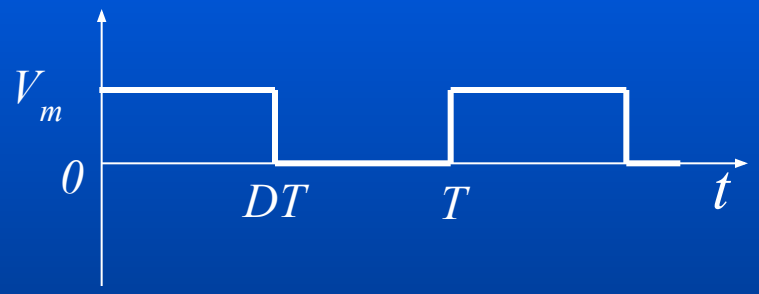
$$p(t) = i^2 R$$

- 직류와 동일한 일을 하는 교류전력값
- 직류전력의 평균값=교류전력의 평균값

$$P = I^2 R = \frac{1}{T}\int_0^T i^2 R(t)\,dt$$

$$I = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt}$$

예제 2-4 펄스파의 평균값



- 듀티비(
=On시간

$$\int 1\,dt = t$$

$$\int x\,dx = \frac{1}{2}x^2$$

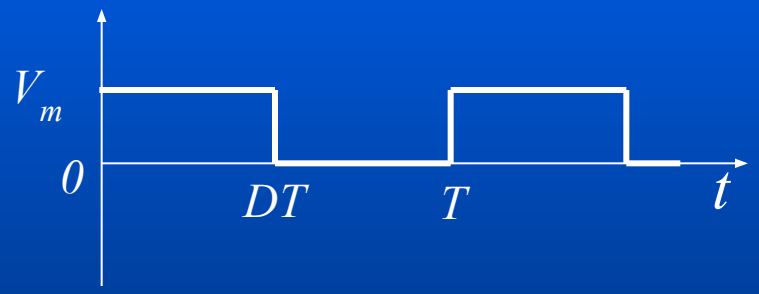
$$\int y^{100}\,dy = \frac{1}{101}y^{101}$$

$$v(t) = \begin{cases} V_m & 0 < t < DT \\ 0 & DT < t < T \end{cases}$$

$$V_0 = \frac{1}{T}\int_0^T v(t)\,dt = \frac{1}{T}\left(\int_0^{DT} V_m\,dt + \int_{DT}^T 0\,dt\right)$$

$$= \frac{V_m}{T}\left[t\right]_0^{DT} = \frac{V_m}{T}\left[DT - 0\right] = V_m D$$

예제 2-4 펄스파의 실효값



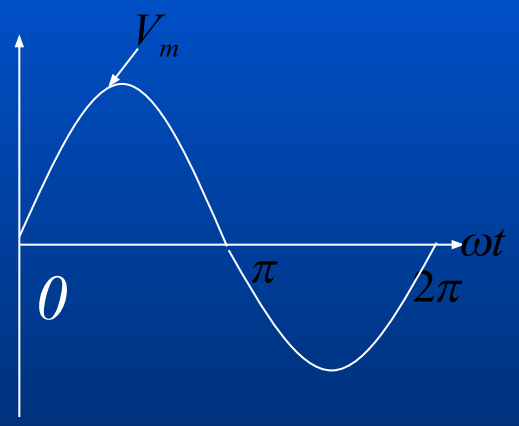
- 듀티비(duty ratio) =On시간/주기(T)

$$v(t) = \begin{cases} V_m & 0 < t < DT \\ 0 & DT < t < T \end{cases}$$

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} = \sqrt{\frac{1}{T}\left(\int_0^{DT} V_m^{\ 2}\,dt + \int_{DT}^{T} 0\,dt\right)}$$

$$= \sqrt{\frac{V_m^{\ 2}}{T}[t]_0^{DT}} = \sqrt{\frac{V_m^{\ 2}}{T}[DT - 0]} = V_m\sqrt{D}$$

예제 2-5 정현파의 평균값



$$V_0 = \frac{1}{\pi}\int_0^{\pi}\left(V_m \sin\omega t\right)d(\omega t)$$

$$= \frac{V_m}{\pi}\left[-\cos\omega t\right]_0^{\pi}$$

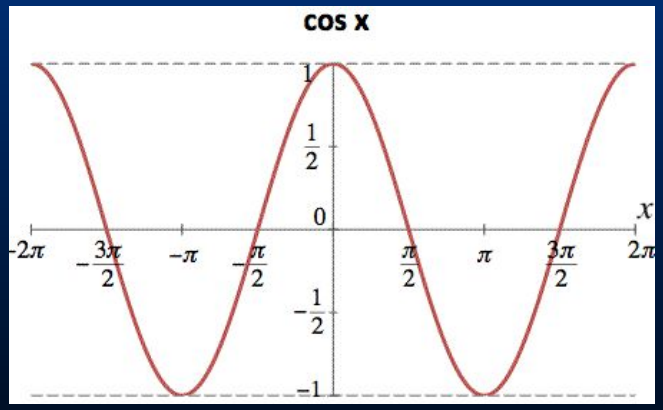
$$\int \sin dt = -\cos t$$

$$= \frac{V_m}{\pi}\left\{-\cos\pi - \left(-\cos 0\right)\right\}$$

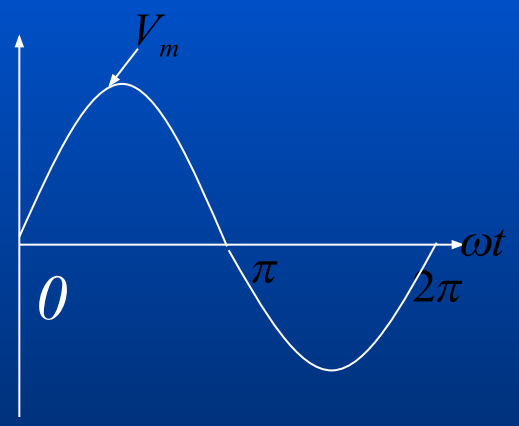
$$= \frac{V_m}{\pi}\left\{1-\left(-1\right)\right\}$$

$$= \frac{2V_m}{\pi}$$

- 사인파는 반주기로 계산
  한주기 계산시 0가 됨



예제 2-5 정현파의 실효값



$$V_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} (V_m \sin\omega t)^2 \, d(\omega t)}$$

$$where\left(\sin^2 x = \frac{1-\cos 2x}{2}\right)$$

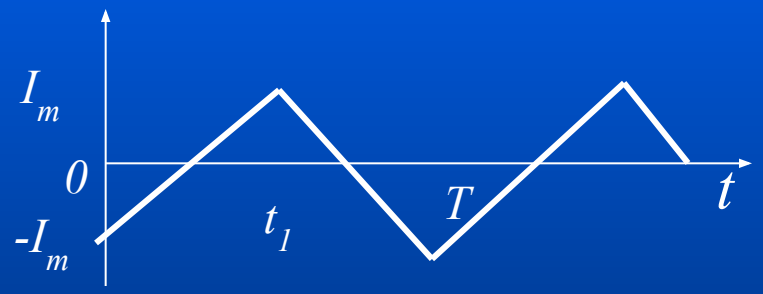
$$= \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi} \frac{1-\cos 2\omega t}{2} d(\omega t)}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_0^{\pi}}$$

$$= \sqrt{\frac{V_m^2}{\pi}\left(\frac{\pi}{2} - \cancel{\frac{\sin 2\pi}{4}} - \frac{0}{2} + \cancel{\frac{\sin 0}{4}}\right)} = \frac{V_m}{\sqrt{2}}$$

- 사인파는 반주기로 계산
  한주기 계산시 0가 됨

예제 2-8 삼각파의 실효값



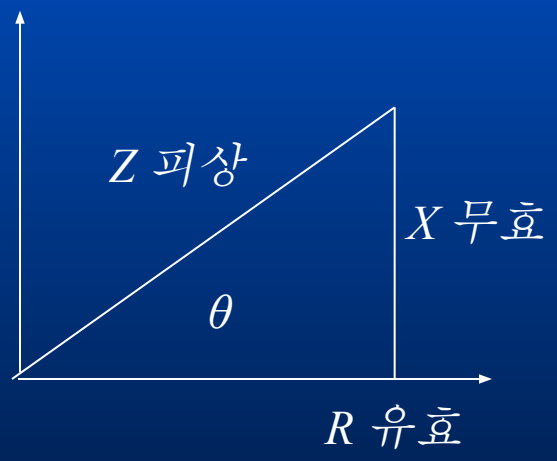
- 듀티비(duty ratio) =온시간/주기(T)

$$i(t)=\begin{cases}\dfrac{2I_m}{t_1}t-I_m & 0<t<t_1\\[2ex] \dfrac{-2I_m}{T-t_1}t+\dfrac{I_m(T+t_1)}{T-t_1} & t_1<t<T\end{cases}$$

$$I_{rms}=\sqrt{\frac{1}{T}\int_0^{t_1}\left(\frac{2I_m}{t_1}t-I_m\right)^2dt}+\sqrt{\frac{1}{T}\int_{t_1}^{T}\left(\frac{-2I_m}{T-t_1}t+\frac{I_m(T+t_1)}{T-t_1}\right)^2dt}$$

$$=\frac{I_m}{\sqrt{3}}$$

2.6 피상전력과 역률



- 피상전력(Apparent power)

$$S = V_{rms} I_{rms} = \sqrt{P^2 + Q^2}$$

- 역률(Power factor)

$$pf = \cos\theta = \frac{p}{S} = \frac{p}{V_{rms} I_{rms}}$$

2.7 정현교류회로의 전력계산(1)

$$v(t) = V_m \cos(\omega t + \theta),\ \ i(t) = I_m \cos(\omega t + \phi)$$

$$p(t) = v(t)i(t) = [V_m \cos(\omega t + \theta)][I_m \cos(\omega t + \phi)]$$

$$\left(\begin{aligned} where\ \cos A \cos B &= \frac{1}{2}[\cos(A+B) + \cos(A-B)] \\ \sin A \sin B &= \frac{1}{2}[-\cos(A+B) + \cos(A-B)] \\ \sin A \cos B &= \frac{1}{2}[\sin(A+B) + \sin(A-B)] \end{aligned} \right)$$

$$p(t) = \frac{V_m I_m}{2}[\cos(\omega t + \theta + \omega t + \phi) + \cos(\omega t + \theta - \omega t - \phi)]$$

$$= \frac{V_m I_m}{2}[\cos(2\omega t + \theta + \phi) + \cos(\theta - \phi)]$$

2.7 정현교류회로의 평균전력계산(2)

$$p(t)=\frac{V_m I_m}{2}\left[\cos(2\omega t+\theta+\phi)+\cos(\theta-\phi)\right]$$

*한주기 적분은 0

$$P=\frac{1}{T}\int_0^T p(t)dt=\frac{V_m I_m}{2T}\int_0^T\left[\cos(2\omega t+\theta+\phi)+\cos(\theta-\phi)\right]dt$$

- 임의소자의 피상전력

$$p(t)=\frac{V_m I_m}{2T}\int_0^T\left[\cos(\theta-\phi)\right]dt=\frac{V_m I_m}{2T}\cos(\theta-\phi)\left[t\right]_0^T$$

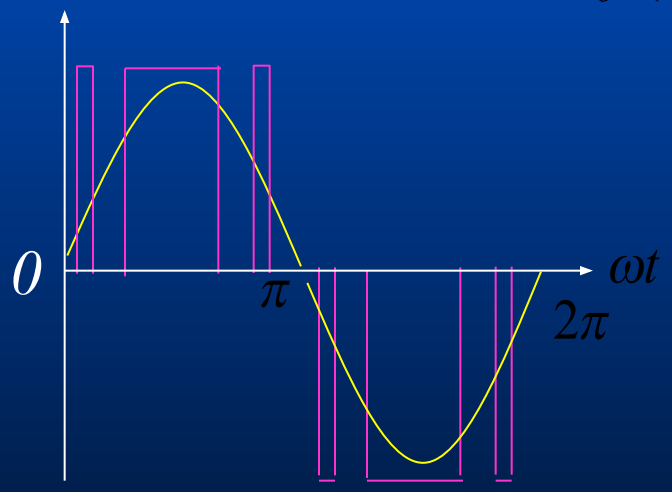
$$=\frac{V_m I_m T}{2T}\cos(\theta-\phi)=\frac{\sqrt{2}V_{rms}\sqrt{2}I_{rms}}{2}\cos(\theta-\phi)$$

$$=V_{rms}I_{rms}\cos(\theta-\phi)$$

2.8 비정현파 주기함수의 전력계산

- Fourier Series(푸리에 급수)

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)\right]$$



$$a_0 = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)dt$$

$$a_n = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\cos(n\omega_0 t)dt$$

$$b_n = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(n\omega_0 t)dt$$

2.8 비정현파 주기함수의 전력계산(2)

- 정현함수로 합치면

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)\right]$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n) \qquad f(t) = a_0 + \sum_{n=1}^{\infty} C_n \sin(n\omega_0 t + \theta_n)$$

$$C_n = \sqrt{a_n^2 + b_n^2} \qquad C_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1}\frac{-b_n}{a_n} \qquad \theta_n = \tan^{-1}\frac{a_n}{b_n}$$

2.8 비정현파 주기함수: 평균전력

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n)$$

$$i(t) = I_0 + \sum_{n=1}^{\infty} I_n \sin(n\omega_0 t + \theta_n)$$

$$P = \frac{1}{T}\int_0^T v(t)i(t)dt$$

$$P = \sum_{n=1}^{\infty} P_n = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,rms} I_{n,rms} \sin(n\omega_0 t + \theta_n)$$

$$P = \sum_{n=1}^{\infty} P_n = V_0 I_0 + \sum_{n=1}^{\infty} \frac{V_{n,\max} I_{n,\max}}{2} \sin(n\omega_0 t + \theta_n)$$

$$V_{n,\max} I_{n,\max} = \left(\sqrt{2}\right)^2 V_{n,rms} I_{n,rms}$$

2.8 비정현파 주기함수: 왜곡율

$$P=\sum_{n=1}^{\infty}P_n=V_0I_0+\sum_{n=1}^{\infty}\frac{V_{n,\max}I_{n,\max}}{2}\sin(n\omega_0 t+\theta_n)$$

$$P=V_0I_0+\frac{V_{1,\max}I_{1,\max}}{2}\sin(\omega_0 t+\theta_1)+\frac{V_{2,\max}I_{2,\max}}{2}\sin(2\omega_0 t+\theta_2)+\cdots$$

- 왜곡율(Distortion factor)

$$DF=\frac{I_{1,rms}}{I_{rms}}$$

- 전고조파왜곡(Total Harmonic Distortion)

$$THD=\sqrt{\frac{\sum_{n\neq 1}I_{n,rms}^2}{I_{1,rms}^2}}=\frac{\sqrt{\sum_{n\neq 1}I_{n,rms}^2}}{I_{1,rms}}=\sqrt{\frac{I_{rms}^2-I_{1,rms}^2}{I_{1,rms}^2}}$$

예제 2.10 정현전원과 비선형 부하

$$v(t) = 100\cos(377t)$$

$$i(t) = 8 + 15\cos(377t + 30) + 6\cos(2.377t + 45) + 2\cos(3.377t + 60)$$

(a) 부하의 소비 전력

$$P = \sum_{n=1}^{\infty} P_n = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,rms} I_{n,rms} \sin(n\omega_0 t + \theta_n)$$

$$= (0)(8) + \left(\frac{100}{\sqrt{2}}\right)\left(\frac{15}{\sqrt{2}}\right)\cos(30^{\circ}) + (0)\left(\frac{6}{\sqrt{2}}\right)\cos(45^{\circ}) + (0)\left(\frac{2}{\sqrt{2}}\right)\cos(60^{\circ})$$

$$= \left(\frac{1500}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = 650[W]$$

예제 2.10 정현전원과 비선형 부하(2)

$$v(t) = 100\cos(377t)$$

$$i(t) = 8 + 15\cos(377t + 30) + 6\cos(2.377t + 45) + 2\cos(3.377t + 60)$$

(b) 부하의 역율

$$V_{rms} = \frac{100}{\sqrt{2}} = 70.7[V]$$

$$I_{rms} = \sqrt{8^2 + \left(\frac{15}{\sqrt{2}}\right)^2 + \left(\frac{6}{\sqrt{2}}\right)^2 + \left(\frac{2}{\sqrt{2}}\right)^2} = 14[A]$$

$$pf = \frac{p}{V_{rms} I_{rms}} = \frac{650}{(70.7)(14.0)} = 0.66$$

예제 2.10 정현전원과 비선형 부하(3)

$$v(t) = 100\cos(377t)$$

$$i(t) = 8 + 15\cos(377t + 30) + 6\cos(2.377t + 45) + 2\cos(3.377t + 60)$$

(c) 부하전류의 왜곡율

$$DF = \frac{I_{1,rms}}{I_{rms}} = \frac{\left(\frac{15}{\sqrt{2}}\right)}{14} = 0.76$$

(d) 부하전류의 전고조파 왜곡율

$$THD = \sqrt{\frac{I_{rms}^2 - I_{1,rms}^2}{I_{1,rms}^2}} = \sqrt{\frac{(14)^2 - \left(\frac{15}{\sqrt{2}}\right)^2}{\left(\frac{15}{\sqrt{2}}\right)^2}} = 0.86$$